

Appendix

I Variational Parameter Updates

In our model, the variational parameters are optimized by maximizing the evidence lower bound (ELBO), defined as:

$$\mathcal{L} = \mathbb{E}_q[\log p(x, z)] - \mathbb{E}_q[\log q(z)] \quad (1)$$

The update rules for the parameters are derived as follows.

Stick-breaking Weights. The Beta distribution parameters in the stick-breaking construction are updated as:

$$\gamma_{k,1} = 1 + \sum_{i=1}^N \phi_{i,k} \quad (2)$$

$$\gamma_{k,2} = \alpha + \sum_{i=1}^N \sum_{j>k} \phi_{i,j} \quad (3)$$

where $\phi_{i,k} = q(z_i = k)$ denotes the variational posterior probability of assigning sample x_i to cluster k , and α is the concentration parameter.

Cluster Counts. The soft count for each cluster c_k is computed as:

$$c_k = c_0 + \sum_{n=1}^N \phi_{n,k} \quad (4)$$

where c_0 is a prior pseudo-count.

Cluster Parameters. The auxiliary count v_k and the cluster center u_k are updated as:

$$v_k = v_0 + \sum_{n=1}^N \phi_{n,k} \quad (5)$$

$$u_k = \frac{1}{c_k} \left(c_0 u_0 + \sum_{n=1}^N \phi_{n,k} f(x_n) \right) \quad (6)$$

where v_0 and u_0 are prior hyperparameters, and $f(x_n)$ denotes the feature embedding of sample x_n .

Inverse Covariance Matrix. The inverse covariance matrix B_k^{-1} of each cluster is updated as:

$$\begin{aligned} B_k^{-1} &= B_0^{-1} + \sum_{n=1}^N \phi_{n,k} \Sigma_n + \sum_{n=1}^N \phi_{n,k} (f(x_n) - \bar{f}_k) (f(x_n) - \bar{f}_k)^\top \\ &\quad + \frac{c_0 N_k}{c_0 + N_k} (\bar{f}_k - u_0) (\bar{f}_k - u_0)^\top \end{aligned} \quad (7)$$

where

$$\bar{f}_k = \frac{1}{N_k} \sum_{n=1}^N \phi_{n,k} f(x_n), \quad N_k = \sum_{n=1}^N \phi_{n,k} \quad (8)$$

Assignment Probabilities. The variational assignment probability $\phi_{n,k}$ is updated in logarithmic form as:

$$\begin{aligned} \log \phi_{n,k} &\propto \Psi(\gamma_{k,1}) - \Psi(\gamma_{k,1} + \gamma_{k,2}) + \sum_{j>k} [\Psi(\gamma_{j,2}) - \Psi(\gamma_{j,1} + \gamma_{j,2})] \\ &\quad - v_k^\top \text{Tr}(B_k \Sigma_n) - v_k^\top f(x_n) - u_k^\top B_k (f(x_n) - u_k) \end{aligned} \quad (9)$$

where $\Psi(\cdot)$ denotes the Digamma function.

II Nonparametric Clustering Method Based on Variational Dirichlet Process (VDP) and Its Advantages Analysis

We adopt the Variational Dirichlet Process (VDP) as the core of our clustering approach to address two key challenges in open-world and long-tailed scenarios: the unknown number of categories and the sparse structure of tail classes. Traditional parametric clustering methods (e.g., DBSCAN, spectral clustering) typically rely on global density assumptions or a predefined number of clusters, making it difficult to effectively capture the complex structures of tail classes in low-density regions, which often results in suboptimal recognition performance.

In the field of nonparametric clustering, alternative methods include Gaussian process-based clustering, hierarchical Bayesian models, and the Chinese Restaurant Process (CRP). Although these methods do not require a preset number of clusters and can adaptively infer clustering structures to some extent, they often suffer from high computational complexity, low inference efficiency, and limited compatibility with

generative models. For example, while CRP offers flexibility in cluster assignments, it lacks explicit modeling of cluster parameters, which limits its ability to accurately characterize tail samples. Similarly, Gaussian process clustering methods face scalability bottlenecks when applied to large-scale datasets, making real-time optimization impractical. Moreover, these approaches typically struggle to guide or integrate with diffusion models, limiting the enhancement of generative quality and diversity based on clustering results.

In contrast, VDP leverages its stick-breaking construction (Equation (1)) and variational inference mechanism to enable both dynamic expansion of cluster numbers and efficient inference. More importantly, VDP incorporates learnable priors, allowing the model not only to adapt to the few-shot and ambiguous boundaries of tail classes (Equations (3)–(4)), but also to provide precise structural guidance for the generative model. Within our integrated clustering-generation framework, VDP is tightly coupled with a diffusion model guided by low-density regions; the clustering results directly participate in the generation process through guidance terms based on cluster weights. This leads to high-quality augmentation of tail-class samples. Such tight integration and unified optimization (Equations (6)–(7), (11)) form a closed-loop synergy between clustering and generation, significantly improving synthesis quality and structural diversity in sparse regions, thereby distinguishing our approach from other nonparametric clustering methods.

We further conduct comparative experiments on ImageNet-100-LT using a range of parametric and nonparametric clustering methods. The detailed results are presented in the following table.

Table 1: Comparison of Clustering Methods on Imagenet-100-LT

Method	Type	Requires Predefined Number of Clusters	New Class Recognition Accuracy (%)
K-means	Parametric	100	38.7
DBSCAN	Parametric	–	41.2
Spectral Clustering	Parametric	100	40.5
Chinese Restaurant Process (CRP)	Nonparametric	–	45.1
Gaussian Process Clustering (GP)	Nonparametric	–	46.3
Variational Dirichlet Process (VDP)	Nonparametric	–	50.2

III VDP-Guided Diffusion Sampling and Long-Tail Optimization

Our method is based on a Variational Dirichlet Process (VDP) guided diffusion model sampling framework, which implements the discretization of the continuous-time reverse diffusion stochastic differential equation (SDE) and dynamic cluster inference and optimization.

The continuous reverse diffusion SDE is formulated as

$$dt = -[\nabla_t \log p(t) + F(t; \theta)]dt + \sigma(t)dW_t, \quad (10)$$

where $\nabla_t \log p(t)$ denotes the score function, $F(t; \theta)$ is the guidance function, and $\sigma(t)dW_t$ is the Wiener process noise. This SDE is discretized using the Euler-Maruyama scheme as

$$t_{i+1} = t_i + \eta_i [\nabla_{t_i} \log p(t_i) + F(t_i; \theta_i)] + \xi_i, \quad (11)$$

where the step size η_i is adaptively adjusted and typically decreases during iterations to ensure numerical stability; the noise term $\xi_i \sim \mathcal{N}(0, \sigma^2(t_i)\eta_i)$ maintains stochasticity consistent with the continuous process variance.

The core of the guidance function $F(t_i; \theta_i)$ is dynamically computed based on the clusters inferred by VDP, defined as

$$F(t_i; \theta_i) = \sum_{k=1}^K w_k(t_i)(\mu_k - t_i), \quad (12)$$

where μ_k is the mean of the k -th cluster, and the posterior weight $w_k(t_i)$ reflects the membership probability of state t_i to cluster k , calculated by

$$w_k(t_i) = \frac{\pi_k \exp\left(-\frac{1}{2}(t_i - \mu_k)^\top \Sigma_k^{-1}(t_i - \mu_k)\right)}{\sum_{j=1}^K \pi_j \exp\left(-\frac{1}{2}(t_i - \mu_j)^\top \Sigma_j^{-1}(t_i - \mu_j)\right)}, \quad (13)$$

where π_k is the cluster weight generated by the stick-breaking process of VDP:

$$\pi_k = \beta_k \prod_{i=1}^{k-1} (1 - \beta_i), \quad \beta_k \sim \text{Beta}(1, \alpha). \quad (14)$$

This design ensures that tail classes with small but nonzero weights are preserved and effectively guide sampling without numerical underflow. In particular, when the covariance matrix Σ_k of a tail class is large, even if t_i is distant from μ_k , the posterior weight $w_k(t_i)$ remains significant, ensuring guidance power in low-density regions.

Appendix

For dynamic cluster management, the model evaluates the Evidence Lower Bound (ELBO) gain to decide whether to introduce new clusters:

$$\Delta\text{ELBO} = \mathbb{E}_{q_{\text{new}}} [\log p(D \mid \theta_{\text{new}})] + D_{KL}(q_{\text{new}} \parallel p) - \mathbb{E}_{q_{\text{exist}}} [\log p(D \mid \theta)], \quad (15)$$

where a positive ΔELBO indicates improved model fit justifying new cluster creation. To control the number of clusters, a merging criterion based on likelihood loss and predictive fit is applied:

$$M(k, \ell) = \alpha - n_k \epsilon \exp(\Delta_{k\ell}) \gamma_{m^*} \mathcal{N}(x_{\text{new}} \mid \mu_{m^*}, \Sigma_{m^*}), \quad (16)$$

where $\Delta_{k\ell} = L_{kk} - L_{k\ell}$ denotes the likelihood drop caused by merging, and ϵ is adaptively adjusted based on sample sizes to balance sensitivity.

The reverse diffusion transition probability at each step is modeled as a Gaussian conditioned on cluster parameters:

$$q(T_{i-1} \mid T_i, \theta_k) = \mathcal{N}\left(T_{i-1}; \mu_k + \Sigma_k(\Sigma_k + \sigma_i^2)^{-1}(T_i - \mu_k), \Sigma_k \sigma_i^2 (\Sigma_k + \sigma_i^2)^{-1}\right), \quad (17)$$

and cluster parameters μ_k, Σ_k are updated online by

$$\mu_k^{(\text{new})} = \frac{n_k \mu_k + \gamma_k t_i}{n_k + \gamma_k}, \quad \Sigma_k^{(\text{new})} = \frac{n_k \Sigma_k + \gamma_k (t_i - \mu_k)(t_i - \mu_k)^\top}{n_k + \gamma_k}, \quad (18)$$

allowing adaptive evolution of the cluster distribution during sampling.

Theoretically, by defining the cluster coverage radius $R_k = \text{tr}(\Sigma_k)$, it can be shown that when the state lies near cluster boundaries, tail classes with larger covariance Σ_k significantly amplify their guidance influence despite small weights π_k , enhancing long-tail sample generation. Furthermore, covariance is adaptively adjusted by

$$\Sigma_k = \Sigma_0 + \beta \log\left(1 + \frac{n_{\max}}{n_k + 1}\right), \quad (19)$$

which enlarges the impact scope of clusters with fewer samples, achieving a variance-adaptive mechanism.

The overall algorithm can be summarized as follows:

Algorithm 1 VDP-Guided Diffusion Sampling

- 1: Initialize sample state $t \leftarrow$ sample noise
 - 2: Initialize clusters using VDP inference
 - 3: **for** each sampling step $i = 1, \dots, T$ **do**
 - 4: Compute posterior weights $w_k = P(z = k \mid t, \text{clusters})$
 - 5: **if** ELBO gain $\Delta\text{ELBO}(t, \text{clusters}) > 0$ **then**
 - 6: Add new cluster with current state t
 - 7: **else if** merge condition satisfied **then**
 - 8: Merge similar clusters to maintain compactness
 - 9: **end if**
 - 10: Compute guidance: $F = \sum_k w_k (\mu_k - t)$
 - 11: Update state:

$$t \leftarrow t + \eta_i (\nabla_t \log p(t) + F) + \xi_i$$
 - 12: Update cluster parameters online based on t
 - 13: **end for**
 - 14: **return** final sample t
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In summary, our method leverages the dynamic cluster inference of VDP and posterior-weighted guidance to effectively protect and enhance tail class generation during reverse diffusion sampling. This substantially improves the model's capability to generate diverse samples under complex, sparse data distributions.

IV Noval Class Recognition

Our model uses cross-entropy loss ($L_{\text{SUP-CE}}$) for known classes, clustering for unknown classes, and balanced losses ($L_{\text{reg}}, L_{\text{proto}}$). Experiments show:

- (a) ResNet + CE fails on novel classes;
- (b) ResNet + clustering separates novel from known classes but not well within novel classes.

Table 2: Experimental results comparing methods on novel accuracy, overall accuracy, and ARI (%).

Method	Novel Acc. (%)	Overall Acc. (%)	ARI (%)
ResNet + CE	9.7	44.2	12.4
ResNet + $L_{\text{ins1}} + L_{\text{ins2}}$	29.7	34.2	31.4
Ours	62.12	57.49	42.9

V Analysis of Overfitting Risk in Tail Classes

We address overfitting caused by limited diversity in class-center sampling of diffusion models through: (1) triple regulation of class attraction, posterior weights, and adaptive noise (Eqs. 6–7) to avoid sample collapse; (2) dual ELBO constraints that jointly optimize quality and diversity; and (3) an adaptive category adjustment (Eq. 15) to dynamically recalibrate generation weights and structure.

To quantify this, we report tail-class Fréchet Inception Distance (FID) scores under varying imbalance factors (IF). Despite expected increases in FID as tail size decreases, our method retains low FID under realistic long-tailed (LT) settings (e.g., IF=50/100), showing good visual quality:

Table 3: Tail-class FID scores under varying imbalance factors (IF). Lower FID indicates better visual quality.

Dataset	IF=10	IF=20	IF=50	IF=100
CIFAR10-LT	2.09	3.23	6.56	9.09
CIFAR100-LT	2.33	4.31	7.56	12.39
ImageNet100-LT	3.56	7.67	12.23	15.32

VI Hyperparameter Sensitivity

Hyperparameters were optimized using Bayesian search over 0, 10, 10, 1. Most λ values are stable in 0.5–0.9, with λ_3 (regularization loss) showing the most impact. λ_1 and λ_5 balance multi-view consistency, while λ_2 – λ_4 handle contrastive, regularization, and clustering losses. Below are sensitivity results (novel class accuracy %): Based on the individual sensitivity analysis of the five hyperparameters, we found that

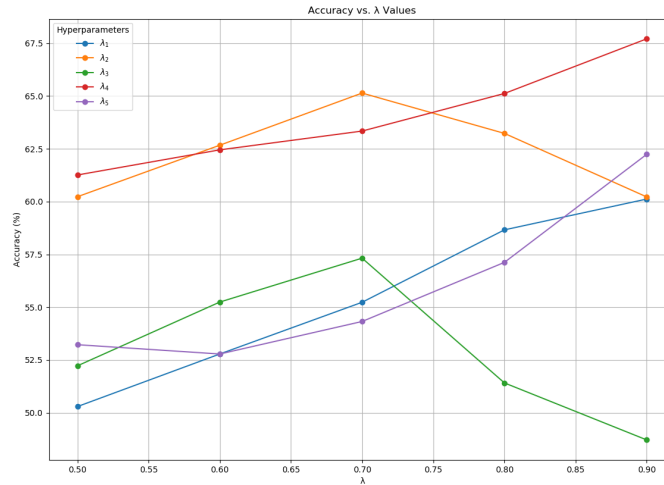


Figure 1: λ hyperparameter sensitivity analysis results (novel class accuracy %)

most parameters ($\lambda_1, \lambda_2, \lambda_4, \lambda_5$) perform stably and achieve higher accuracy within the range of 0.7 to 0.9. In contrast, the regularization loss weight λ_3 is more sensitive to performance, with the optimal range between 0.6 and 0.7; deviating from this range significantly reduces accuracy. The results are shown in Figure 1

VII More Imbalanced Datasets

To bridge the scale gap, we additionally include experiments on the downsampled ImageNet-100-LT, which provides a mid-scale long-tailed setting with 100 classes. In subsequent work, we will further expand to datasets with more diverse imbalance patterns, including both synthetic and real-world data sources, to comprehensively evaluate the scalability and applicability of our framework.

The table below reports results on CIFAR-100-LT under varying imbalance factors ($\rho = 20, 50, 100$). Our method consistently achieves strong performance across seen and novel classes, especially under high imbalance.

Table 4: Performance (%) on CIFAR-100-LT under different imbalance factors (ρ).

Method	$\rho = 20$			$\rho = 50$			$\rho = 100$		
	Seen	Novel	Overall	Seen	Novel	Overall	Seen	Novel	Overall
TRSSL	61.82	33.94	56.54	52.62	28.45	48.31	47.51	26.13	45.81
ORCA	56.61	27.43	50.71	50.04	25.35	45.83	45.12	20.24	42.12
GPC	45.54	42.52	37.32	40.21	40.21	32.82	35.38	30.01	32.13
BaCon-O	61.21	57.62	59.71	53.42	58.44	56.21	56.22	57.12	55.81
Ours	62.12	57.43	59.37	54.93	55.42	56.23	58.81	54.91	56.42